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PALAEOMAGNETIC DIRECTIONS IN PRECAMBRIAN BASIC INTRUSIVES OF THE MOUNT ISA PROVINCE, AUSTRALIA

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Palaeomagnetic investigation of basic intrusives in the Proterozoic Mount Isa Province yields three groups of directions of stable components of NRM after magnetic cleaning in fields up to 50 mT (1 mT = 10 Oe) The youngest group (IA) includes results from the Lakeview Dolerite, and yields a palaeomagnetic pole at 12°S, 124°E (A_{95} = 11°) The second group (IB) has a palaeomagnetic pole 53°S, 102°E (A_{95} = 11°) The third group (IC) is derived from the Lunch Creek Gabbro and contains normal and reversed polarities of magnetization with a palaeomagnetic pole at 63°S 201°E (A_{95} = 9°) Some samples from the gabbro have anomalously low intensities of remanent magnetization in obscure directions attributed to the relative enhancement of the non-dipole component of the palaeomagnetic field during polarity reversal The present attitude of the igneous lamination is probably of primary, not tectonic origin

1 Introduction

Palaeomagnetic studies carried out on separate shield regions within continents can potentially provide a unique test for large-scale relative motion between cratons, enabling the nature of continental accretion during the Precambrian to be investigated If such studies were to show that relative movements have been small (as has been claimed for Africa by Piper et al [1] and others) then this would be critical to assessing the role of the plate-tectonic concept for the Precambrian

In this paper palaeomagnetic results from dolerite dykes and a lopolith (Lunch Creek Gabbro) from the Proterozoic Mount Isa Geosyncline [2] of northwest Queensland are presented Similar work has recently been carried out on other Precambrian cratons in Australia [3–5] The geological implications of the results from the Lunch Creek Gabbro in the "eastern succession" of the geosyncline are also discussed For instance it is possible using palaeomagnetic methods to clarify the significance of the dip of igneous lamination in this intrusive, and to place a constraint on the minimum cooling time for the body

2 Geological outline

2 1 Geological setting and age relationships

At an early stage in its evolution, the region was divided into two distinct depositional basins separated

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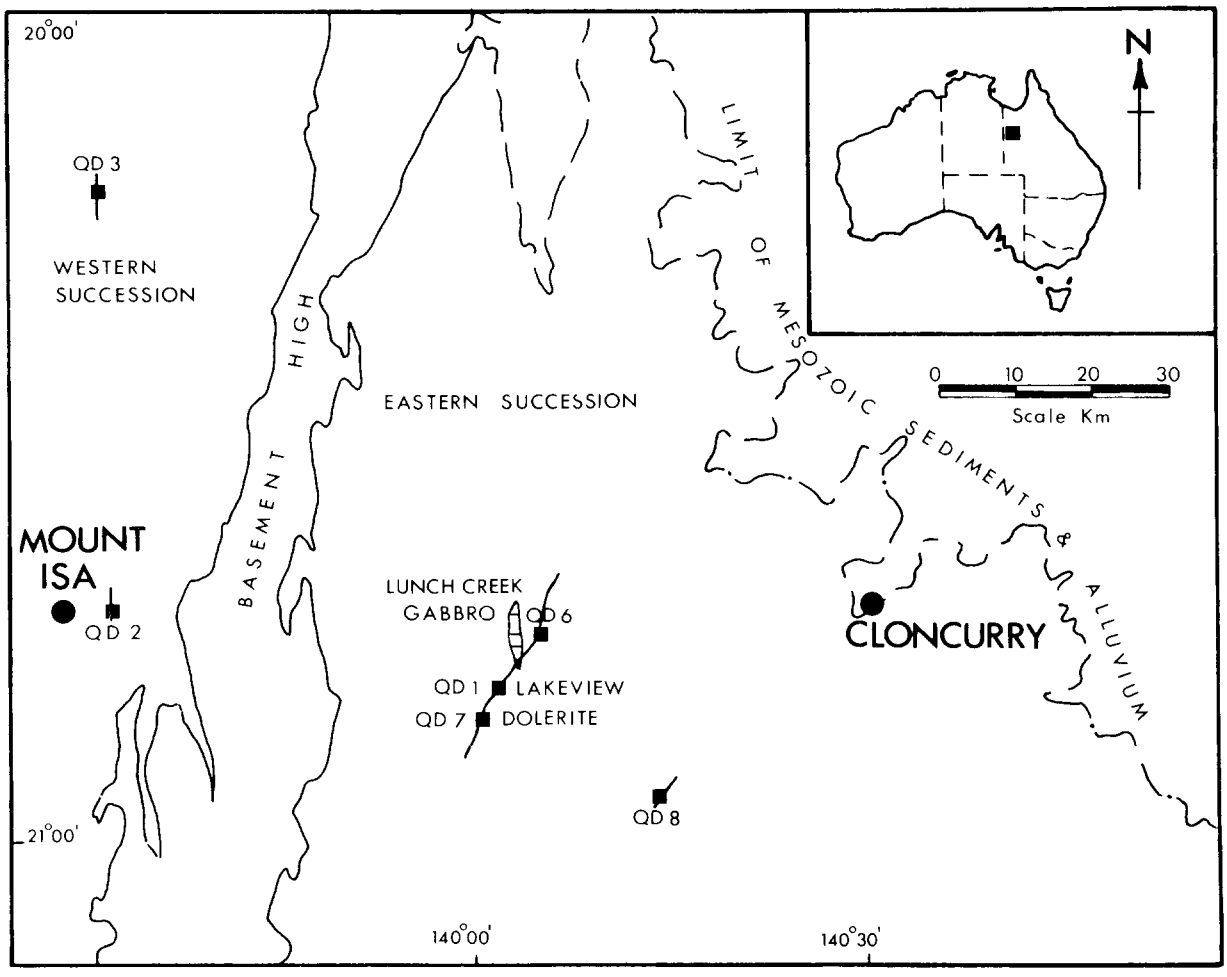


Fig 1 Sampling localities in the Mount Isa–Cloncurry Province The dyke trends are indicated schematically with solid lines

by a meridional basement "high" [2] shown in Fig 1 The oldest rocks are basement metamorphics dated at around 1710 m y [6] Whilst granitic intrusion occurred intermittently throughout the period 1700–1400 m y , predominantly in the eastern succession, basic dykes were emplaced into both the eastern and western successions in discrete episodes [2,7]

The Lunch Creek Gabbro [7] is a lopolith about 10 km long (elongate north–south) intruding the Corella Formation, a carbonate-pelite-arenite sequence of the eastern succession It is itself intruded by the synmetamorphic Burstall Granite, dated between 1410 m y and 1470 m y [6], and by the younger Lakeview Dolerite [7] (Fig 2)

Regional metamorphism affected both gabbro and Corella Formation at about 1400–1450 m y [8] Little is known of the time relationships of the dykes sampled in the western succession They are younger than 1700 m y and probably younger than the regional metamorphism of about 1450 m y

2 2 Structure and opaque mineralogy

Both successions were folded on north-trending axes producing elongate basins and domes [2] Further deformation was represented by three periods of faulting The Lunch Creek Gabbro, however, appears to have escaped penetrative deformation, and no struc-

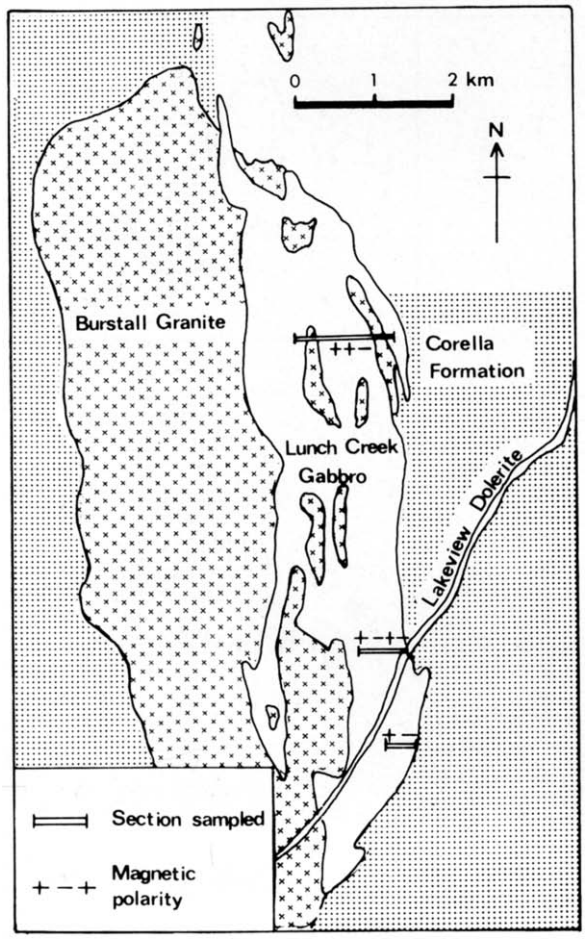


Fig 2 Geology of the Lunch Creek Gabbro indicating location of sections sampled and their corresponding polarity sequences of remanent magnetization

tural fabric can be recognized in it The recognition of folding within the Lunch Creek Gabbro itself is critical to the interpretation of the palaeomagnetic results, directions of primary remanent magnetization must be restored, if necessary, to their original attitude by unfolding the layering [9] The implication of the palaeomagnetic results for the presence of fold structures will be discussed in section 4 The igneous layering is of variable dip and orientation through the intrusion, but typically quite steep The layering results from the segregation of ferromagnesian and quartzo-feldspathic minerals

The principal opaque phase in the Lakeview Dolerite and Lunch Creek Gabbro is magnetite or titanomagnetite up to about 1 mm in size and present as homogeneous grains or rarely with exsolutons of ilmenite and spinel Pyrite and chalcopyrite are commonly present, with rare magnetite inclusions in the pyrite Ilmenite generally forms discrete homogeneous grains

2 3 Sampling

Oriented hand samples and drill cores were obtained from six localities in the youngest set of basic intrusives Two north-trending gabbroic dykes, about 100 m wide and intruding metabasalt, were sampled in the western succession (sites QD2 and QD3) The remaining sites were located in the eastern succession Sites QD1, QD6, and QD7 are separate localities along the northeast-trending Lakeview Dolerite, which is about 30 km long and, at the sampling localities, averages 50 m wide The dolerite dyke sampled at QD8 is about 200 m wide and is concordant with local structural trends

Three east–west traverses were made through separate regions of the Lunch Creek Gabbro (Fig 2), and a total of 42 samples were collected at six sites oriented with both sun and magnetic compass In several places, it forms continuous pavement in creek beds so that within-site distributions of samples also tend to traverse the gabbro

Up to 7 specimens were cut from each of 40 samples from the dykes, and 42 samples from the Lunch Creek Gabbro

3 Palaeomagnetic results

3 1 NRM and AF demagnetization

The directions and intensities of the natural remanent magnetization (NRM) were measured using a P A R spinner magnetometer Although within-sample scatter was generally small, within-dyke directions displayed little consistency Directions from dyke QD8 were close to the direction of the present field axis, and samples from the Lunch Creek Gabbro displayed a planar distribution through the present field axis Samples from the remaining sites were magnetized oblique to the direction of the present field axis with apparently no preferred orientation

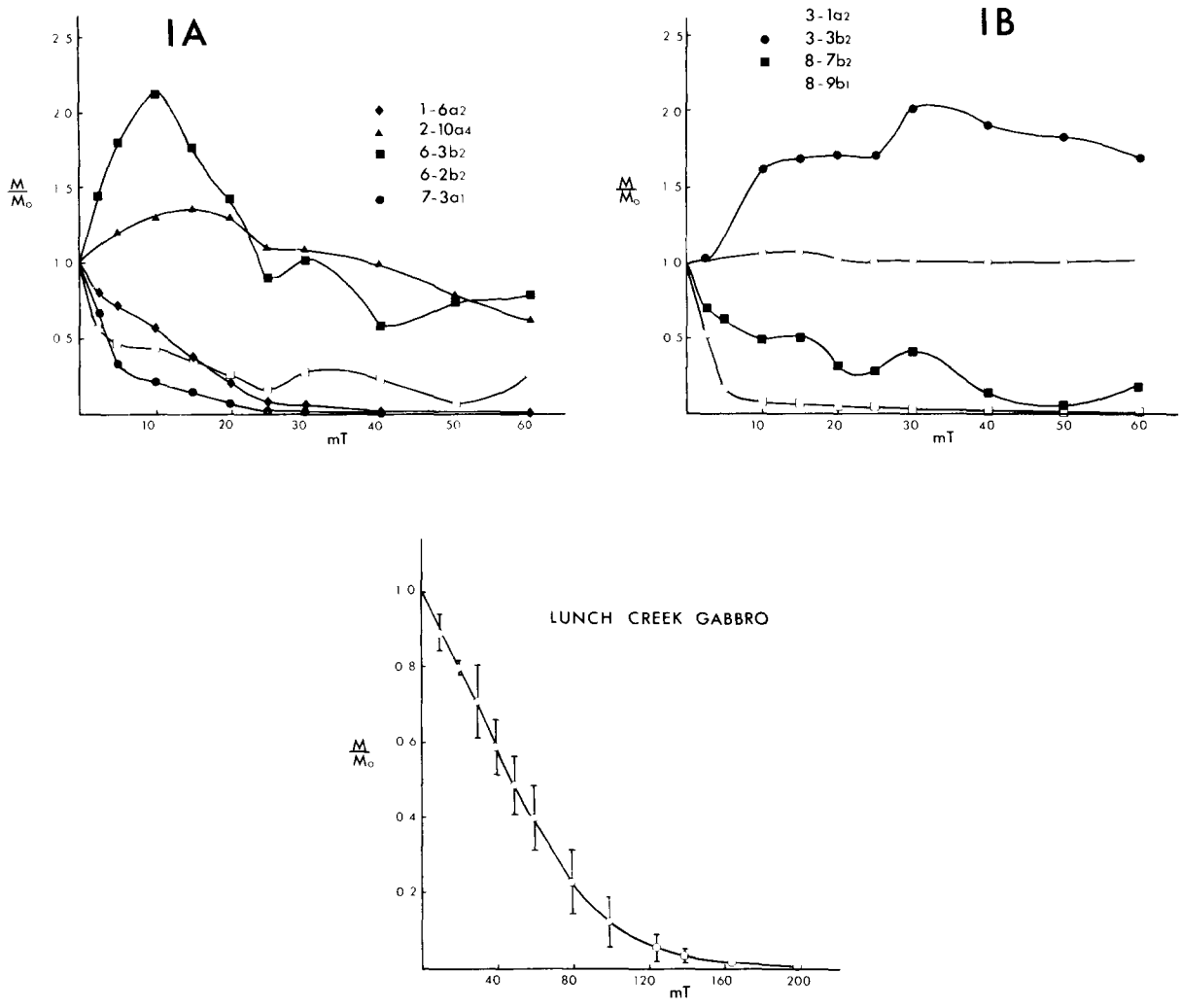


Fig 3 Examples of demagnetization-intensity curves for samples comprising the three groups of directions The standard error of the mean-normalized intensity is shown on the demagnetization curve for the Lunch Creek Gabbro

The stability of NRM was tested in the laboratory with the alternating field (AF) demagnetization technique Twenty-five pilot specimens, chosen from separate samples from all dyke sites and three of the lopolith sites, were treated in incrementally increasing fields to a maximum of 210 mT (peak value) Partial demagnetization in fields of 25–75 mT was required before readings were obtained from four samples from site QD1, during sampling, it was suspected that the Lakeview Dolerite in this region had been struck by lightning thereby acquiring a strong component of isothermal remanent magnetization Some of the demagnetization characteristics are presented in Fig 3 A few specimens show an initial increase in intensity, whereas others display a rapid decrease during the early stages of magnetic cleaning The tendency for the intensities of some specimens to increase after treatment in low fields may be explained by the selective removal of small opposing components of viscous remanent magnetization Specimens from the Lunch Creek Gabbro, however, showed a smooth decrease of intensity during magnetic cleaning An average curve for this gabbro is shown in Fig 3

We have used the method devised by Giddings and

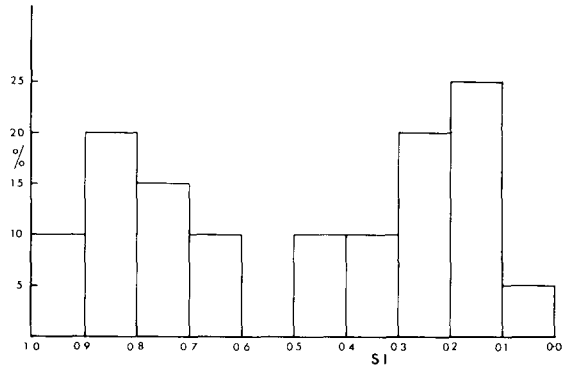


Fig 4 A histogram showing the spectrum of stability indices calculated from measurements on 25 pilot specimens The method follows that described by Giddings and McElhinny [10] The distribution is bimodal, specimens which yield the lower range of values belong to the Lakeview Dolerite (sites QD1, 6 and 7)

McElhinny [10] to define the spectrum of stability indices (SI values) of remanent magnetization measured in the Mount Isa intrusives The index is based on defining (a) the proportion of the coercivity spectrum over which secondary components dominate, (b) the stable range of alternating magnetic fields over which the isolated component is directionally consistent, and (c) measuring the maximum alternating field a single specimen can tolerate Fig 4 shows the SI values for the Mount Isa intrusives are bimodally distributed Values greater than 0 1 are obtained from specimens whose directions are repeatable over a particular range (the stable range) of treatment in alternating magnetic fields The specimens with SI values between 0 and 0 5 belong to sites QD1, 6 and 7 located in the Lakeview Dolerite This provides some evidence that the magnetic phases may be more homogeneously distributed than the demagnetization-intensity curves imply (Fig 3, IA)

Stable components of magnetization from sites QD1, 2, 3, 6, 7 and 8 were isolated after treatment at 20 mT (peak field), cleaned group-mean directions are presented in Table 1 In almost all cases within-sample scatter was reduced by cleaning and within-dyke sample-means grouped oblique to the direction of the present field (Fig 5) Sample-mean directions measured from the Lunch Creek Gabbro converged to two anti-parallel directions after treatment in a peak-alternating field of 50 mT As the ferromagnetic phase is titanomagnetite, which is known to become magnetized parallel to the ambient field, the gabbro has evidently recorded a field reversal Together with the stability in direction and intensity of the remanent magnetization at fields above 50 mT, this is confirmation that the direction of remanent magnetization is of primary origin [11] In unaltered igneous rocks, this is probably a thermoremanence (TRM)

TABLE 1

Group mean directions of cleaned remanent magnetization and corresponding palaeomagnetic pole positions

Rock description	Group	N	R	D	I	Palaeomagnetic pole		A_{95}
						lat	long	
Lakeview Dolerite	IA *	18	17 488	297	+80	12°S	124°E	11°
Gabbroic dykes in the eastern and western successions	IB *	14	13 506	208	+60	53°S	102°E	11°
Lunch Creek Gabbro	IC **	6	5 942	152	+50	63°S	201°E	9°
	IC	42	38 556	149	+52	61°S	200°E	6°

* IA = QD1, 6 and 7, and QD2, IB = QD3 and QD8

** Site-mean data for the Lunch Creek Gabbro

N = total number of samples measured in each group (except IC **), R = resultant of N unit vectors, D = declination, degrees E of N, I = inclination, positive downward, A_{95} = semi-angle of the cone of confidence about the mean of virtual geomagnetic poles at the 95% probability level (analysed following the method of Fisher [12] reversing all directions with negative polarity)

TABLI 2

Comparison of Lunch Creek Gabbro group-mean statistics in situ and relative to the igneous lamination

	R	k	α_{95}	D	I
6 sites in situ	5 942	86	7	152	+50
6 sites relative to igneous lamination	4 653	3 7	40	183	+27
42 samples in situ	38 556	11 9	6	149	+51

k and α_{95} = precision and semi-angle of the cone of confidence at the 95% probability level of the mean direction of remanent magnetization, R, D and I as in Table 1

Sampling details and a Fisher analysis [12] of within-group and within-site distributions are given in Tables 1 and 2

3 2 Mean directions of remanent magnetization and palaeomagnetic pole positions

The sample-mean directions obtained from the four dykes have been separated into two groups (Fig 5, Table 1) comprising

(1) The Lakeview Dolerite (QD1, 6 and 7) from the eastern succession, and dyke QD2 from the western succession – group IA The mean palaeomagnetic pole position is situated at 12°S, 124° (A_{95} = 11°)

(2) Dykes QD3 and QD8 from the western and eastern successions respectively – group IB, with a mean palaeomagnetic pole position at 53°S, 102°E (A_{95} = 11°)

Sites within the Lunch Creek Gabbro yield a third group of directions – IC, see Fig. 5 – with a corresponding pole position at 63°S, 201°E (A_{95} = 9°), regarded as a true palaeomagnetic pole

The data, although preliminary, indicate that by combining the results from palaeomagnetism and geochronology it may eventually be possible to establish a correlation between periods of dyke intrusion in the eastern and western successions The age relationship of the pole for dyke QD3 in group IB with that for dyke QD2 is equivocal, the disparate directions for these dykes may suggest a protracted period of north-trending basic dyke intrusion in the western succession and/or rapid apparent polar wandering during the period of intrusion

Interpretation of the data in terms of apparent polar wandering during the Precambrian will only be possible when comparison is made with results from other cratons currently being investigated However, palaeolatitudes suggested from group IA are consistent with those expected for the Adelaidean of Australia, indicators from rocks of this age suggest the prevalence of polar conditions [13] The intermediate palaeolatitudes calculated for the Lunch Creek Gabbro are consistent with recent Carpentarian palaeomagnetic results from West Australian rocks of comparable age [3]

4 Lunch Creek Gabbro

4 1 Field reversal

4 1 1 Minimum cooling for the Lunch Creek Gabbro

As the remanence is believed to be stable TRM and dual polarity is observed, it follows that if the duration of geomagnetic field reversal is known, then this provides a minimum estimate of the time taken for the gabbro to cool below 580°C, the Curie temperature of magnetite Recent estimates of times for polarity transitions recorded in volcanic rocks and deep-sea sediments of Recent and Tertiary age [14–17] vary from about 1000 to 4000 years, but a period of about 2000 years is frequently noted If Precambrian polarity transitions were of similar duration – and this has not been conclusively established (though they were similar to modern cases in every other respect) – then an estimate of 2000 years for the minimum cooling time may apply to the Lunch Creek Gabbro

4 1 2 Anomalous remanence directions and intensities

The 8 sample-mean directions most remote from the estimated mean of the positive and negative populations (Fig 5d) have intensities of NRM commonly one to two orders of magnitude lower than the other samples Moreover, these samples were all collected from localities separating zones of positive and negative polarity in the gabbro (Fig 2) These observations together are consistent with a model of geomagnetic field reversal involving a reduction in the field strength

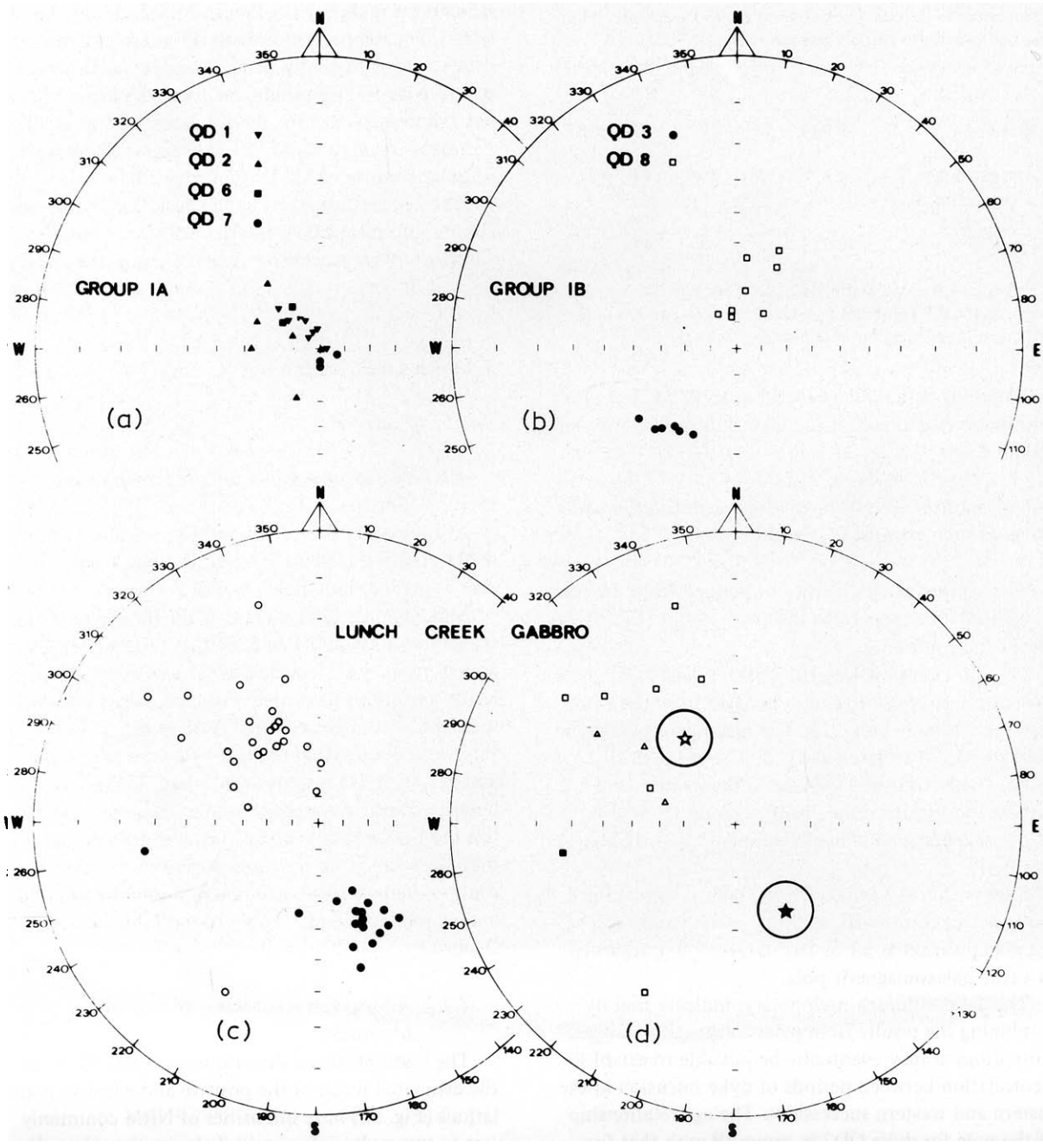


Fig 5 (a, b) Sample-mean directions of magnetization from the basic dykes obtained after cleaning (20 mT) (c) Sample-mean directions of magnetization from the Lunch Creek Gabbro obtained after cleaning (50 mT) (d) Directions from Lunch Creek Gabbro samples with very low intensities of NRM (□) and of samples from the baked contact (△) showing their relation to the estimated mean direction with its circle of 95% confidence (Equal angle projection, positive directions, closed symbols, negative directions, open symbols)

and obscure changes of field direction during polarity transition, as has been observed in the case of other field reversals by numerous authors [18–20] The usual explanation is that the non-dipole component of the field remains essentially constant in strength while the main dipole field decays

4 1 3 Polarity zones within the gabbro

In a cooling lopolith it might be expected that the Curie isotherm would migrate slowly upward and toward the centre of intrusion [21], and if reversals of the geomagnetic field occur during this period, this movement should be recorded in the sequence of polarities observed in the body [22] For an irregularly shaped intrusion such as the Lunch Creek Gabbro many observations scattered throughout the complex are required in order to map this change However, a polarity sequence that is consistent from west to east can be recognized in three regions where traverses were made The sequence recognized, + – + –, can be generated in two main ways (Fig 6)

(1) By folding and/or faulting of the initial igneous layering Unless all folding occurred above the Curie temperature (implying synmetamorphic deformation with temperatures in excess of 580°C), this is virtually precluded by the strong grouping of remanent magnetization directions in antiparallel positive and negative populations

(2) By the cooling of the gabbro through two periods of geomagnetic reversal

Whichever alternative is correct, a very slow cooling rate for the lopolith is implied

Three gabbro samples were taken from the baked contact of the Lunch Creek Gabbro within 1 m of the Burstall Granite, in a positive polarity zone in the gabbro They yield negative directions of remanent magnetization similar to the negative direction in the gabbro complex itself (Fig 5d) This suggests that intrusion of this phase of the Burstall Granite into the Lunch Creek Gabbro may have occurred very shortly after cooling of the gabbro, and implies an age of about 1470 m y for the Lunch Creek Gabbro [6]

4 2 Significance of the tilted igneous layering

In Table 2 site-mean statistics are calculated both in situ and relative to the local plane of igneous lamination in order to provide a "fold test" [9] The in situ grouping is much the more precise (k = 86) and the drastic drop in precision (k = 3 7) upon adjustment to the lamination strongly suggests that the dip of this lamination predates the stable remanence (that is, it was acquired above the Curie temperature of the ferromagnetic minerals) It is, however, beyond the capability of magnetic investigation to resolve whether this dip was "depositional", imposed during the inferred long initial cooling history of the complex, or during any subsequent synmetamorphic deformation

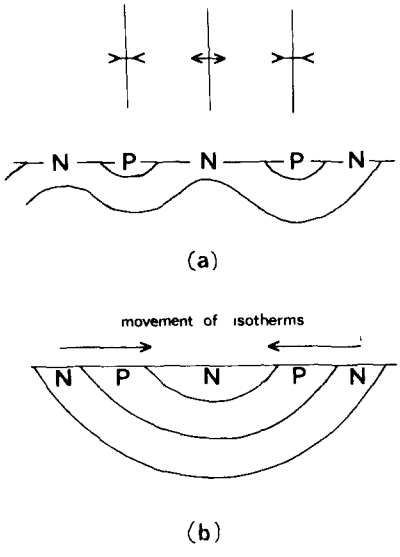


Fig 6 Alternative explanations of the reversal sequence, + – + –, in east–west sections through the Lunch Creek Gabbro (a) Repetition of an initial – + sequence by folding of the igneous layering (b) Cooling of the lopolith through two periods of geomagnetic field reversal

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